Chapter 1 Riemann Surfaces

1.1 Definitions

Definition 1.1.0.1 A **Riemann surface** is one-dimensional complex=connected 1-dimensional complex analytic manifold.

That is, equipped with maximal charts $\{(U_\alpha, z_\alpha)\}_{\alpha\in A}$ on M

$$z_\alpha : U_\alpha \to \mathbb{C}$$

Then,if transition functions are holomorphic

$$f_{\alpha\beta} = z_\alpha \circ z_\beta^{-1} \; :; U_\alpha \cap U_\beta = \varnothing$$

The charts are called **holomorphic compactible**.

◇ **Lemma 1.1.0.2** (Local-Global Principle) For every family $\{V_k\}$ of open sets, and union $V = \cup V_k$: A function $f : V \to \mathbb{C}$ is holomorphic$\Leftrightarrow$ $f|V_k$ is holomorphic.

Definition 1.1.0.3 Classically, a compact Riemann surface is called **closed**, elsewise is **open**.

1.1.1 Complex Structure

Definition 1.1.1.1

- A (complex) atlas is the collection of pair
$$\{(U_\alpha, z_\alpha)\}$$
- Two atlas $\mathfrak{U}, \mathfrak{U}'$ are **analytically equivalent** if every chart of $\mathfrak{U}'$ is holomorphically compatible with every chart of $\mathfrak{U}$ is holomorphically compatible with every chart of $\mathfrak{U}'$.
- **complex structure** is an equivalence class of complex structure.

Tori

Definition 1.1.1.2 Suppose $\omega_1, \omega_2 \in \mathbb{C}$ are linearly independent over $\mathbb{R}$, then

$$\Gamma := \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$$

is the **lattice** spanned by ω_1, ω_2. The set of all equivalence classes is denoted by

$$\mathbb{C}/\Gamma$$

is the tori.

The topology of Tori: Induced by the quotient topology

...

◇ **Lemma 1.1.1.3** $\mathbb{C}/\Gamma$ is a Riemann surface.

∘ **Proof** Induced by quotient topology, the corresponding atlas. □

◇ **Lemma 1.1.1.4** (Universal Covering of Tori) The mapping

$$\pi : \mathbb{C} \to \mathbb{C}/\Gamma$$

is a universal covering.

∘ **Proof** By definition, $\mathbb{C}$ is simply connected, locally path-connected, semi-locally path connected.
And, it is a covering map: Select some $\bar{z} \in \mathbb{C}/\Gamma$, then select a neighborhood $\bar{U}$, then $\pi^{-1}(\bar{U}) = \cup_{m,n\in\mathbb{Z}} U + (n\omega_1 + m\omega_2)$.
Now, if U is small enough, $= \sqcup_{m,n\in\mathbb{Z}} U + (n\omega_1 + m\omega_2)$. This is possible by Hausdorffness. □

♠ **Proposition 1.1.1.5** Let $\Gamma = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2, \Gamma' = \mathbb{Z}\omega_1' + \mathbb{Z}\omega_2'$, then the followings are equivalent:

- $\Gamma = \Gamma'$;
- There exists a matrix $A \in SL_2(\mathbb{Z})$ such that

$$(\omega_1, \omega_2)^T = A(\omega_1', \omega_2')$$

∘ **Proof** It is easy to verify: If $\Gamma = \Gamma'$, then when we obtain ω_1, ω_2 as $\mathbb{Z}$-vector,

$$\omega_1 = \lambda_1\omega_1' + \lambda_2\omega_2'$$
$$\omega_2 = \mu_1\omega_1' + \mu_2\omega_2'$$

□

♠ **Proposition 1.1.1.6** Suppose: Γ, Γ' be two lattice, with $\alpha \in \mathbb{C}^*$ such that $\alpha\Gamma \subseteq \Gamma'$.
Then, the map

$$z \mapsto \alpha z \qquad \mathbb{C} \to \mathbb{C}$$

induces a holomorphic mapping $\mathbb{C}/\Gamma \to \mathbb{C}/\Gamma'$.
Then, it is biholomorphic, iff $\alpha\Gamma = \Gamma'$.

∘ **Proof** ⇐ It is easy to check;
⇒ If $\mathbb{C}/\Gamma, \mathbb{C}/\Gamma'$ are biholomorphic, then we lift the map to unviersal covering:

$$\begin{array}{ccc} \mathbb{C} & \overset{\tilde{f}}{\dashrightarrow} & \mathbb{C} \\ \downarrow & & \downarrow \\ \mathbb{C}/\Gamma & \xrightarrow{f} & \mathbb{C}/\Gamma' \end{array}$$

And such $\hat{f}$ is unique, given by $z \mapsto \alpha z$. We ibtain that $\Gamma = \alpha\Gamma'$. □

Especially, a special art of tori is given by the lattice

$$\Gamma = \mathbb{Z} + \mathbb{Z}\tau, \tau \in \mathbb{C} \qquad Im(\tau) > 0$$

◇ **Lemma 1.1.1.7** Every tori $\mathbb{C}/\Gamma$ is biholomorphic to a torus of the form

$$X(\tau) := \mathbb{C}/(\mathbb{Z} + \mathbb{Z}\tau)$$

∘ **Proof** Suppose $\Gamma = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2, arg(\omega_2) > arg(\omega_1)$, we just use the spin $z \mapsto re^{i\theta}z$, where

$$r = |\frac{1}{\omega_1}| \qquad \theta = -arg(\omega_1)$$

then we get a lattice $\mathbb{Z} + \mathbb{Z}\tau$. Furthermore, we could calculate τ:

$$\tau = |\frac{\omega_2}{\omega_1}|e^{arg(\omega_2)-\arg(\omega_1)} = \frac{\omega_2}{\omega_1}$$

□

Then, under base change of Γ:

$$\begin{cases} \omega_1' = a\omega_1 + b\omega_2 \\ \omega_2' = c\omega_1 + d\omega_2 \end{cases}, \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$$

We obtain that,

$$\tau' = \frac{\omega_1'}{\omega_2'} = \frac{a\omega_1 + b\omega_2}{c\omega_1 + d\omega_2}$$

Thus:

♠ **Proposition 1.1.1.8** Suppose $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z}), Im(\tau) > 0$, then for

$$\tau' = \frac{a\tau + b}{c\tau + d}$$

$X(\tau)$ is biholomorphic to $X(\tau')$.

∘ **Proof** Omitted. □

1.1.2 Holomorphic Functions

Definition 1.1.2.1 Let X be a Riemann surface, $Y \subseteq U$ be open subset, then $f : Y \to \mathbb{C}$ is called **holomorphic**, if

$$f \circ \psi^{-1} : \psi(U \cap Y) \to \mathbb{C}$$

is holomorphic, for every chart (U, ψ). Such collection can be denoted by $\mathcal{O}(Y)$.

But, this lead to a question: Is the complex structure of a Riemann surface unique? The fact is, no.

♠ **Proposition 1.1.2.2** (The $\mathbb{C}$-Algebra Structure) $\mathcal{O}(X)$ is a $\mathbb{C}$-algebra,

∘ **Proof** By conclusion of $\mathbb{C}$-algebra of $\mathcal{O}(U), U \subseteq \mathbb{C}$. □

▶ ***Theorem*** (Riemann, Removable Singularity) If f is holomorphic on $U - \{a\}$, and is bounded in some neighborhood of a, then f can be uniquely extended to $\tilde{f}$.

∘ **Proof** By the statement in complex ana. □

Definition 1.1.2.3 Suppose X, Y be Riemann surface. Then, the function $f : X \to Y$ is called **holomorphic** if

$$\varphi \circ f \circ \psi^{-1} : \psi(U \cap X) \to \varphi(V \cap X)$$

is holomorphic.

Especially, when $Y = \mathbb{C}$, take the chart of $(\mathbb{C}, id_{\mathbb{C}})$.

▶ ***Theorem*** (Identity Theorem) Suppose X, Y be Riemann surfaces, and $f_1, f_2 : X \to Y$ are two holomorphic mappings, such that

- $f_1|A = f_2|A$;
- A has a limit point/accumulation poitn $a \in X$

Then, $f_1 = f_2$.

∘ **Proof** ... □

▶ ***Theorem*** ...

∘ **Proof** ... □

Mobius Transforamation

♠ **Proposition 1.1.2.4** ...

∘ **Proof** ... □

1.1.3 Meormorphic Functions, Function Field

Definition 1.1.3.1 ...

1.1.4 Local Properties

Local Behavior, Open Mapping

▶ ***Theorem*** (Local Behavior of Holomorphic Mappings) Suppose X, Y be Riemann surfaces, and $f : X \to Y$ be a non-constant holomorphic mapping.

Suppose $a \in X$, $b = f(a)$, then there exists an integer $k \geq 1$ and chart $\varphi : U \to V$ on X,$\psi : U' \to V'$ on Y with following properties:

- $a \in U, \varphi(a) = 0; b \in U', \psi(b) = 0$;

- $f(U) \subseteq U'$;
- $F = \psi \circ f \circ \varphi^{-1} : V \to V'$ is given by

$$F(z) = z^k \qquad \forall z \in V$$

∘ **Proof** Notice that, for arbitrary atlas, $\psi \circ f \circ \varphi^{-1}$ is holomorphic, and apply the conclusion in complex analysis. We obtain

$$F(\xi) = (\xi h(\xi))^m \qquad g(\xi) = e^{L(\xi)}, h(\xi) = e^{\frac{L(\xi)}{m}}$$

Now, instead of φ^{-1}, we use a local coordinate by compositition with $\phi : \xi :\mapsto \xi h(\xi)$, then we obtain

$$F \circ \phi^{-1}(\eta) = \eta^m$$

□

★ **Corollary 1.1.4.1** Let X, Y be Riemann surfaces, and let $f : X \to Y$ be a nonconstant holomorphic mapping. Then, f is open, i.e. the image of open subset is open.

∘ **Proof** Apply the conclusion. □

★ **Corollary 1.1.4.2** Let X, Y be Riemann surfaces, and $f : X \to Y$ be an injective holomorphic mapping. Then, f is biholomorphic mapping of X onto $f(X)$.

∘ **Proof** Then, the degree $m = 1$. Thus, the inverse mapping is holomorphic. □

Definition 1.1.4.3 If $w = z^{e_p}$, then $e_p \geq 1$ is called **ramification index**.
And the **branching number** is defined by $b_f(p) := e_p - 1$.

Maximum,Minimum Principle

★ **Corollary 1.1.4.4** If $f : X \to \mathbb{C}$ is a nonconstant holomorphic function, then the absolute value of f, $|f|$ not attain its maximum.

∘ **Proof** Notice $|f|$ is also open. □

▶ ***Theorem*** Let M, N be Riemann surface, with M compact, then for holomorphic $f : M \to N$, f is either:

- Constant; or
- Surjective.

In particular, $\mathcal{O}(M) = \mathbb{C}$.

∘ **Proof** Firstly, f is open. And M is compact. Thus, if f is nonconstant, $f(M)$ is open and closed, $f(M) = N$. □

★ **Corollary 1.1.4.5** Every holomorphic function on a compact Riemann surface is constant.

∘ **Proof** Notice that, $\mathbb{C}$ is not compact, but f is open. □

★ **Corollary 1.1.4.6** Every meromorphic function f on ^{1} is rationnal. i.e. it can be written as quotient of two polynomials.
i.e. $\mathbb{C}(z) = \mathcal{O}(^1)$

∘ **Proof** The poles:
If f has infinity numbers of poles, then it has a limit point(by compactness), by identity theorem,

$$f = \infty$$

If f has finite many of poles, then we could choose a chart by Mobius transformation

$$z \mapsto \frac{1}{z-a}$$

Then, suppose the poles are in the affine chart $\mathbb{C} =^1 -\{\infty\}$, then

$$f = g + \sum h_v$$

$g \in \mathcal{O}(P^1) \Rightarrow g = c$. Thus, f is rational. □

Liouville's Theorem

▶ ***Theorem*** Every bounded holomorphic function on $\mathbb{C}$ is constant.

∘ **Proof** If f is bounded, then by Riemann's removable singularity, it can be extended to holomorphic $^1 \to \mathbb{C}$, thus is constant. □

1.1.5 Double Periodic Functions

Definition 1.1.5.1 Suppose $\omega_1, \omega_2 \in \mathbb{C}$, are linearly independent over $\mathbb{R}$, with Γ be the lattice.
Then, a meromorphic function $f : \mathbb{C} \to ^1$ is called **doubly periodic** with respect to Γ, if

$$f(z) = f(z+\omega), \forall z \in \mathbb{C}, \omega \in \Gamma$$

▶ ***Theorem*** Every doubly periodic holomorphic function $f : \mathbb{C} \to \mathbb{C}$ is constant.
Every non-constant periodic meromomrphic function $f : \mathbb{C} \to ^1$ attains every value $c \in ^1$.

∘ **Proof**

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□

Example: Weierstrass's ζ-Function

1.2 Topology of Riemann Surface

1.2.1 Fundamental Groups

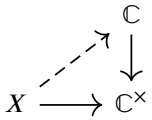
Given in topology.

1.2.2 Covering Manifolds

1.2.3 Branched and Unbrached Covering

Logarithm

♠ **Proposition 1.2.3.1** $exp : \mathbb{C} \to \mathbb{C}^\times$ is a universal covering. Thus, for every simply-connected Riemann surface X, the following diagram commutes



Especially, if $X = \mathbb{C}^\times$, $f = id$, then the lifting is the logarithm.

∘ **Proof** $\mathbb{C}$ satisfies those properties of universal covering. Then, we obtain the lifting. □

Ramification/Branched Point

Definition 1.2.3.2 f is called **ramified** or **branched** at p if the ramification number

$$F(z) = z^k \qquad b_p(f) = k > 1$$

Elsewise is **uniramified** or **unbranched**.

♠ **Proposition 1.2.3.3** (Criteria)

∘ **Proof** ... □

1.2.4 Euler Characteristics of Surface, and Riemann-Hurwitz Relation

Recall: The **Euler-Poincare charater** of a compact surface of genus g is described as

$$\chi = \alpha_0 - \alpha_1 + \alpha_2$$

Recall The structure of compact orientable surface:

◇ **Lemma 1.2.4.1** For surface M, N,

$$\chi(M \# N) = \chi(M) + \chi(N) - 2$$

▶ ***Theorem*** The genus of surface is defined as:

$$M \cong \#_g T^2 = \Sigma_g$$

Thus, $H_1(M) \cong \mathbb{Z}^{2g}$.

★ **Corollary 1.2.4.2** We obtain, that

$$\chi(\Sigma_g) = 2 - 2g$$

Definition 1.2.4.3 For $f : X \to Y$, the **Branching number** is defined as

$$B(f) := \sum_{P \in M} b_f(P)$$

To show, $B(f)$ is well-defined:

◇ **Lemma 1.2.4.4** If $f : X \to Y$ is a non-constant holomorphic mapping, then the ramification set is discrete.

∘ **Proof** Notice that, $df_p \neq 0, \forall p \in X$, then the zero of f' is discrete. Thus, the ramification set is discrete. □

◇ **Lemma 1.2.4.5** If $f : X \to Y$ is a holomorphic mapping, then the ramification set is finite

∘ **Proof** The discrete subset is compact Hausdorff space should be finite. □

▶ ***Theorem*** (Riemann-Hurwitz) We have

$$g = n(\gamma - 1) + 1 + \frac{B}{2}$$

∘ **Proof** Firstly, the surface is can be triangulated with the finite subset

$$S = \{f(x) | b_f(x) > 0\}$$

By F faces, E edges, V vertices.

Lift this to M. The induced triangulation of M has: nF faces, nE edges, $nV - B$ vertices. Because: The inside of triangles contains no branch point.

We obtain, that

$$\begin{cases} F - E + V = 2 - 2\gamma \\ nF - nE + nV - B = 2 - 2g \end{cases}$$

And solve the equation □

★ **Corollary 1.2.4.6** Branching number B is always even.

∘ **Proof** Omitted. □

1.2.5 Cohomology of Riemann Surface

1.2.6 Poincare's Lemma in Complex:Dolbeault's Lemma

1.2.7 Serre's Duality

▶ ***Theorem*** (Serre)

We will prove this in complex geometry.

Chapter 2 Analytic Continuation, Covering Surfaces and Algebraic Functions

2.1 Analytic Continuation

Definition 2.1.0.1

- (Function Element) Let M, N be fixed Riemann surfaces. A **function element** is pair

$$(f, D)$$

  where:

  - D is a connected, open nonempty subset of M;
  - $f : D \to N$ is analytic.

- Two elements $(f_1, D_1), (f_2, D_2)$ are **direct analytic continuation** of each other if

$$D_1 \cap D_2 \neq \emptyset \qquad f_1|D_1 \cap D_2 = f_2|D_1 \cap D_2$$

  Use the following notion $(f_1, D_1) \simeq (f_2, D_2)$.

Definition 2.1.0.2 (Analytic Continuations)

- We say $(f, D), (g, \tilde{D})$ are **analytic continuation** if there exists function elements $(f_j, D_j); 1 \leq j \leq J$:

$$(f_1, D_1) = (f, D) \qquad (f_J, D_J) = (g, \tilde{J}) \qquad (f_j, D_j) \simeq (f_{j+1}, D_{j+1})$$

- A **compelete analytic continuation** is an equivalence class of suc (f, D). By analytic continuation.

◇ **Lemma 2.1.0.3** (Analytic Continuation⇒ Path) Suppose $(f, D), (g, \tilde{D})$ is an analytic continuation. Then, there exists a path

$$\gamma : [0, 1] \to M$$

Connecting the points of D, with following property: Exists

$$0 < t_1 < \cdots < t_m = 1$$

where: Exists $B(\gamma(t_i), r_i)$

∘ **Proof** ... □

★ **Corollary 2.1.0.4** ...

∘ **Proof** ... □

2.2 The Unramified Riemann Surface of An Analytic Germ

2.3 The Ramified Riemann Surface of An Analytic Germ

2.4 Algebraic Germs and Functions

Chapter 3 Analysis on Riemann Surface

3.1 Differential Forms and Integration Formula

3.1.1 Complex Structure

Definition 3.1.1.1 Let X be a Riemann surface.
Then, the **(almost)complex structure** of X is defined as following:

$$J : TM \to TM \qquad J_p^2 = -id_{T_p}$$

Then, $(0,1)$ and $(1,0)$-form are specified by

$$T^{*(1,0)}X := Ker(J^* - i) \qquad T^{*(0,1)}X := Ker(J^* + i)$$

Remark In higher dimension, there is a difference between complex structure and almost complex structure, depends on the integrability.

♠ **Proposition 3.1.1.2** Suppose X be a Riemann surface, then, the complexification of cotangent space satisfies

$$T^*X \otimes_{\mathbb{R}} \mathbb{C} = T^{*(1,0)}X \oplus T^{*(0,1)}X$$

where TR^*X is the real cotangent space.

∘ **Proof** Notice: $dim_{\mathbb{R}}(T_pM) = 2$, thus

$$ker(J_p^* + i) \oplus ker(J_p^* - i) = T_pM$$

□

3.1.2 Differential Forms

Definition 3.1.2.1 For a Riemann surface M:

- The 0-form is the function on M;
- The 1-form is the form such that

  $$f\,dx + g\,dy; f, g \in C(U, \mathbb{C})$$

  is invariant under coordinate change.
- The 2-form is the form

  $$f\,dx \wedge dy; f \in C(U; \mathbb{C})$$

  which is also invariant under coordinate change.

Take the following notion of $\Omega^k(M; \mathbb{C})$.

i.e. For the coordinate change $\tilde{z}$, we obtain the new 1-form $\tilde{f}(\tilde{z}), \tilde{g}(\tilde{z})$, with $\tilde{\omega} = \tilde{f}d\tilde{x} + \tilde{g}d\tilde{y}$:

$$\begin{pmatrix} d\tilde{x} & d\tilde{y} \end{pmatrix} \begin{pmatrix} \tilde{f}(\tilde{z}) \\ \tilde{g}(\tilde{z}) \end{pmatrix} = \begin{pmatrix} dx & dy \end{pmatrix} \begin{pmatrix} \frac{\partial \tilde{x}}{\partial x} & \frac{\partial \tilde{y}}{\partial x} \\ \frac{\partial \tilde{y}}{\partial x} & \frac{\partial \tilde{y}}{\partial y} \end{pmatrix} \begin{pmatrix} \tilde{f}(\tilde{z}) \\ \tilde{g}(\tilde{z}) \end{pmatrix} \Rightarrow \begin{pmatrix} \tilde{f}(\tilde{z}) \\ \tilde{g}(\tilde{z}) \end{pmatrix} = \begin{pmatrix} \frac{\partial \tilde{x}}{\partial x} & \frac{\partial \tilde{y}}{\partial x} \\ \frac{\partial \tilde{y}}{\partial y} & \frac{\partial \tilde{y}}{\partial y} \end{pmatrix} \begin{pmatrix} \tilde{f}(\tilde{z}) \\ \tilde{g}(\tilde{z}) \end{pmatrix}$$

and the new 2-form $\tilde{\omega} = \tilde{f} d\tilde{x} \wedge d\tilde{y}$

$$\tilde{\omega} = \tilde{f}(\frac{\partial \tilde{x}}{\partial x} + \frac{\partial \tilde{x}}{\partial y})dx \wedge (\frac{\partial \tilde{y}}{\partial x} + \frac{\partial \tilde{y}}{\partial y})dy \Rightarrow \tilde{f}(\tilde{z}) = f(z(\tilde{z}))\frac{\partial(x,y)}{\partial(\tilde{x},\tilde{y})} = |\frac{\partial z}{\partial \tilde{z}}|^2 f(z(\tilde{z}))$$

where $\frac{\partial(x,y)}{\partial(\tilde{x},\tilde{y})}$ is the determinant of Jacobian.

♠ **Proposition 3.1.2.2** We obtain, that

$$dz \wedge d\bar{z} = -2idx \wedge dy$$

∘ **Proof** $(dx + idy) \wedge (dx - idy) = -2idx \wedge dy$ □

♠ **Proposition 3.1.2.3** We obtain: For a Riemann surface M,

$$\Omega^k(M;\mathbb{C}) = \Omega^k(M;\mathbb{R}) \otimes_{\mathbb{R}} \mathbb{C}$$

∘ **Proof** Notice: We decompose the differential form into

$$f dx + g dy = (u_1 + iv_1)dx + (u_2 + iv_2)dy = (u_1 dx + u_2 dy) + i(v_1 dx + v_2 dy)$$
$$f dx \wedge dy = (u + iv)dx \wedge dy = f dx \wedge dy + i(v dx \wedge dy)$$

□

And in the viewpoint of differential, there exists pairing

$$\Omega^1(M;\mathbb{C}) \times TM \to \mathbb{R} \qquad (\omega, v) \mapsto \omega(v)$$

Then, with the decomposition of TM, we obtain that

★ **Corollary 3.1.2.4** $\Omega^1(M;\mathbb{C}) = \Omega^{(1,0)}(M) \oplus \Omega^{(0,1)}(M)$, where

$$\Omega^{(1,0)}(M) = \cap_{v \in T^{0,1}(M)} Ker(< -, v >) = Ann(T^{(0,1)M})$$
$$\Omega^{(0,1)}(M) = \cap_{v \in T^{1,0}(M)} Ker(< -, v >) = Ann(T^{(1,0)M})$$

∘ **Proof** Easy to verify. □

The differential form could be written in a conjugation of differentials

$$f dx + g dy = u dz + v d\bar{z} \qquad \begin{cases} f = u + v \\ g = i(u - v) \end{cases} \qquad \begin{cases} dz = dx + idy \\ d\bar{z} = dx - idy \end{cases}$$

Similarly, the 2-form

$$g(z)dz \wedge d\bar{z} = -2ig(z)dx \wedge dy$$

Definition 3.1.2.5 The **conjugation** of differential 1-form

$$\omega = f dx + g dy \Rightarrow \bar{\omega} = \bar{f} dx + \bar{g} dy$$

Integration

Definition 3.1.2.6

- (Line Integration) Given $\gamma : I \to M, \omega \in \Omega^1(M)$, then

$$\int_\gamma \omega := \int_0^1 (f(x,y)\frac{dx}{dt} + g(x,y)\frac{dy}{dt})dt$$

- (Area Integration) Given area D with $\omega \in \Omega^2(M)$, then

$$\iint_D \Omega = \iint_D f(x,y)dx \wedge dy$$

Differential of C^1 Form

For C^1-Form $\omega \in \Omega^1(M;\mathbb{C})$

$$d\omega = f\,dx + g\,dy \Rightarrow d\omega = (g_x - f_y)dx \wedge dy$$

Thus, the differential is finally a 2-form. Now, apply Stokes's theorem

...

where:

Conjugation of Differentials

Definition 3.1.2.7

- Let $f \in C^1(U,\mathbb{C})$, then define

$$\partial f := f_z dz \qquad \bar{\partial} f = f_{\bar{z}} dz$$

where the notion $f_z, f_{\bar{z}}$ are Wirtinger derivation.

- For the differential form $\omega = u dz + v d\bar{z}$,

$$\partial\omega = v_z dz \wedge d\bar{z}$$
$$\bar{\partial}\omega = -u_{\bar{z}} dz \wedge d\bar{z}$$

where

$$f_z = \frac{1}{2}(f_x - i f_y) \qquad f_{\bar{z}} = \frac{1}{2}(f_x + i f_y)$$

Notice: $f_{\bar{z}} = 0 \Leftrightarrow f$ is holomorphic.

♠ **Proposition 3.1.2.8** We obtain a decomposition

$$d = \partial + \bar{\partial}$$

And

$$\partial^2 = \partial\bar{\partial} + \bar{\partial}\partial = \bar{\partial}^2$$

∘ **Proof** ... □

Under the form $\omega = f\,dz + g\,d\bar{z}$, we obtain

$$\bar{\omega} =$$

Hodge Star Operator

Definition 3.1.2.9 Given: A Riemann surface M, with a non-vanishing 2-form $\lambda dx \wedge dy$ on the surface(canonical 2-form), and

- A function f, then

$$* f = f(z)(\lambda(z)dz \wedge dy)$$

- A differential 1-form

$$\omega = f dx + g dy \Rightarrow * \omega := -g dx + f dy$$

  which is called **Hodge star** of 1-form ω.

- A differential 2-form

$$* \Omega = \Omega / \lambda(z) dx \wedge dy$$

Except the 1-form, it depends on the conformal metric.

♠ **Proposition 3.1.2.10** If we take the complex notion $\omega = u dz + v d\bar{z}$, then

$$* \omega = -iu dz + iv d\bar{z}$$

∘ **Proof** $* \omega = -g dx + f dy = i(u - v)\frac{dz + d\bar{z}}{2} + (u + v)\frac{dz - d\bar{z}}{2i} = \frac{i}{2}[(u - v)(dz + d\bar{z}) - (u + v)(dz - d\bar{z})]$ $i(-ud z + v d\bar{z})$. □

3.1.3 Exactness, Closedness

Definition 3.1.3.1 Let $\omega \in \Omega^1(M; \mathbb{C})$.

- ω is called **exact** if

$$\omega = df \qquad f \in C^2$$

  And coexact if $* \omega$ is exact;

- ω is called **closed** if it is C^1, and

$$d\omega = 0$$

  And cocolosed if $* \omega$ is exact;

Laplacian,Harmonic Differentials

Definition 3.1.3.2 Suppose $f \in (U, \mathbb{C})$, then define the **Laplacian**

$$\Delta f = (f_{xx} + f_{yy}) dx \wedge dy$$

Especially, a function is called **harmonic** if $\Delta f = 0$.

♠ **Proposition 3.1.3.3** For $f \in C^2(U)$,

$$\Delta f = -2i\bar{\partial}\partial f = d * df$$

∘ **Proof** Notice:

$$-2i\bar{\partial}\partial f = -2i\bar{\partial}(f_z dz) = -2i\bar{\partial}(\frac{1}{2}(f_x - if_y)dz) = \frac{i}{2}(f_{xx} + if_{xy} - if_{yx} + f_{yy})dz \wedge d\bar{z} = \frac{1}{2i}(f_{xx} + f_{yy})dz \wedge d\bar{z} = (f_{xx} + f_{yy})dx \wedge$$

$$d * df = d * (f_x dx + f_y dy) = d(-f_y dx + f_x dy) = -(f_{xy} dx \wedge dx + f_{yy} dy \wedge dx) + (f_{xx} dx \wedge dy + f_{yx} dy \wedge dy) = (f_{xx} + f_{yy})dx$$

□

Definition 3.1.3.4 1-form ω is called **harmonic** if $\omega = df$ locally, $\Delta f = 0$;

♠ **Proposition 3.1.3.5** A differential ω is harmonic, iff it is closed and co-closed.

∘ **Proof** $\Rightarrow$: Notice that $d(*\,\omega) = 0$. And it is closed becasue $d^2 f = 0$;
$\Leftarrow$: Notice $\omega = df$ locally, and $d(*\,df) = 0$. Thus, f is harmonic. □

Definition 3.1.3.6 Take the following notion:

$$\mathcal{H}^k(M;\mathbb{C}) := \{\omega \in \Omega^k(M) | \Delta\omega = 0\}$$

3.1.4 n-Homology Group

Pairing Between n-form and n-Homology Group

Definition 3.1.4.1 Define the following integral pairing

$$H_n(X;\mathbb{Z}) \times Z^n(X;\mathbb{C}) \to \mathbb{C} \qquad ([c],\omega) \mapsto \int_c \omega$$

Thus, we obtain the equivalence

...

3.1.5 Holomorphic 1-Form

Definition 3.1.5.1 A 1-form ω is called **holomorphic** provided that locally:

$$\omega = df \qquad f \in \mathcal{O}(U)$$

♠ **Proposition 3.1.5.2** If u is a harmonic function on M, then ∂u is a holomorphic differential.

∘ **Proof** Notice: $\Delta u =$ □

A special case is the complex $\mathbb{C}$. We obtain $\Delta f = 0 \Rightarrow f \in \mathcal{O}(U)$.

▶ ***Theorem*** A differential form ω is **holomorphic** iff

$$\omega = \alpha + i * \alpha$$

for some harmonic differential α;

∘ **Proof** ... □

★ **Corollary 3.1.5.3** A differential ω is holomorphic iff it is closed, and $* \omega = -i\omega$.

∘ **Proof** ... □

3.2 Integration Formula

▶ ***Theorem*** (Integration by Parts) Let D be a relative compact region on a Riemann surface M, with piecewise differentiable boundary.

Let $f \in C^1(M)$ and ω be a C^1 form on a neighborhood of the closure of D, then

$$\int_{\partial D} f\omega = \iint f d\omega - \iint_D \omega \wedge df$$

∘ **Proof** ... □

★ **Corollary 3.2.0.1** If ω is closed, in particular or holomorphic 1-form

$$\int_{\partial D} \omega = 0$$

∘ **Proof** ... □

★ **Corollary 3.2.0.2** ...

∘ **Proof** ... □

3.3 L^2 Structure of Riemann Surface

3.3.1 Weyl's Theorem

3.3.2 L^2 Space

3.4 Mittag-Leffers Obstruction

Chapter 4 Geometry of Riemann Surface

4.1 Geometric Properties

4.1.1 Riemann-Metric Structure

4.1.2 Orientability

♠ **Proposition 4.1.2.1** Every Riemann surface is orientable.

∘ **Proof** Notice that, for □

Conversely:

▶ ***Theorem*** Every orientable connected surface admits a complex structure.

∘ **Proof** ... □

4.2 Line Bundles

Definition 4.2.0.1 Let X be an algebraic curve. A **line bundle** over X is

$$\pi : L \to X$$

where for every $p \in X$, there exists $p \in U \subseteq X$ such that $\pi^{-1}(U) \cong U \times \mathbb{C}$.

This implies, that we obtain a projection map

$$p : \sqcup U_p \times \mathbb{C} \to X$$

Factorization

$$\sqcup U_p \times \mathbb{C} \to L \to X$$

Definition 4.2.0.2 Such U with $\pi^{-1}(U) \cong U \times \mathbb{C}$ is called **support** of line bundle.
Equippedd with $\phi : \pi^{-1}(U) \to U \times \mathbb{C}$.

Thus, for $pr_1 : U \times \mathbb{C} \to U$, we obtain that

$$pr_1 \circ \phi = \pi | \pi^{-1}(U) : \pi^{-1}(U) \to U$$

Definition 4.2.0.3 Let X

Homomorphisms

Invertible Sheaves

Rational Section

4.2.1 Holomorphic Line Bundle

4.2.2 The Tautological Line Bundle

4.2.3 Canonical Bundle

4.2.4 Tangent Bundle

4.2.5 Picard Group

Chapter 5 Compact Riemann Surface

5.1 Compact Riemann Surface

5.1.1 Differential Forms

Residue

Definition 5.1.1.1 Let f be a (local) meromorphic function, $\omega = f\,dz$ be a meromorphic 1-form. Then, $\omega = (\sum_{-m}^{\infty} a_m z^m)dz$, and define

$$Res_p(\omega) := a_{-1}$$

◇ **Lemma 5.1.1.2** (Local Integration) Suppose $p \in Pole(\omega) \subseteq M$, then for a small enough closed curve around p, γ_p,

$$\oint_{\gamma_p} \omega = 2\pi i Res_p(\omega)$$

Here: Small enough means, we obtain a local coordinate in a small enough unit disk:

$$U \ni p, z : U \to \Delta$$

such that $z(p) = 0$(in center). And U contains no other pole. $Im(\gamma_p) \subseteq U$.

∘ **Proof** Notice that

$$\oint_{\gamma_p} \omega = \oint_{\gamma_p} (\sum_{-m}^{\infty} a_m z^m)dz$$

Now, by Cauchy's integration formula

$$\overset{z=\epsilon e^{i\theta}}{=} \int_0^{2\pi} (\sum_{-m}^{\infty} a_m (\epsilon e^{i\theta})^m)\epsilon i e^{i\theta}$$
$$= \sum_{-m}^{\infty}$$

□

Global Residue Theorem

▶ ***Theorem*** (Global Residue Theorem) Given a meromorphic function $f \in M(C)$, C is a compact Riemann surface, then

$$\sum_p Res_p(\frac{df}{f}) = 0$$

∘ **Proof** ... □

★ **Corollary 5.1.1.3** (Argument Principle)

∘ **Proof** ... □

5.1.2 Meromorphic Functions

♠ **Proposition 5.1.2.1** Let $f \in M(C)$, C is a compact Riemann surface, then define:Number of poles $n(f)$ and number of zeros $m(f)$:

$$n(f) = m(f)$$

∘ **Proof** A viewpoint is to observe $f : C \to \mathbb{C}$ as a holomorphic map

$$\tilde{f} : C \to^1 \mathbb{C}$$

Then, $n(f) = card(f^{-1}(\infty))$, $m(f) = card(f^{-1}(0))$, we obtain that $n(f) = m(f)$
Another way is by: Suppose f be locally in the form of

...

□

5.1.3 Decompositon of Differential Form and Cohomology

Hodge Decomposition of Forms, Hodge Theorem

▶ ***Theorem*** (Hodge) For a compact Riemann Surface M,

$$\mathcal{H}^k(M) = H^k_{dR}(M)$$

∘ **Proof** ... □

Hodge Decomposition of Cohomology

5.1.4 Harmonic Differentials

5.1.5 Analytic Differentials

5.2 Intersection Theory

5.2.1 Intersection Number

Topological Definition

Definition 5.2.1.1 Let a, b be two cycles on the Riemann surface M. We define the **intersection number**

$$a * b := \int_M \eta_a \wedge \eta_b$$

where $\eta_s = PD([a])$ by poincare duality.

Geometric Definition

Definition 5.2.1.2 Let M be oriented, and γ_1, γ_2 be two oriented closed curves, and tranverse $T_p\gamma_1 \oplus T_p\gamma_2 = T_pM$(Equivalently, not tangent)
Then,

$$\gamma_1 \cdot \gamma_2 := \sum_{p \in \gamma_1 \cap \gamma_2} \epsilon_p \qquad \epsilon_p = \Big\{ ...$$

5.2.2 Local Intersection

Recall the contents in algebraic curve.

5.3 Riemann-Roch Theorem

5.3.1 Divisors

Definition 5.3.1.1 Suppose C be a Riemann surface, then the **divisor** is the finite sum

$$D = \sum m_i p_i \qquad m_i \in \mathbb{Z}, p_j \in C$$
$$deg(D) := \sum m_i \qquad Div(C) \to \mathbb{Z}$$

Especially, for the meromorphic function $f \in K(C)^\times = M(C)$, then

$$(f) = \sum_{p \in C} v_p(f)p$$

v_p is the degree of f at p.

This implies

♠ **Proposition 5.3.1.2**

∘ **Proof** ... □

$$f : C \to^1 \mathbb{C}$$

▶ ***Theorem*** If $f \in K(C)$ is not constant, and C is compact, then

$$deg(f) = \sum v_p(f) = 0$$

∘ **Proof** Notice that: On compact Riemann surface, the number of poles is equal to zeros. □

Bezout's Theorem

▶ ***Theorem*** (Bezout)

∘ **Proof** ... □

Ramification Divisors

5.3.2 Riemann-Roch Theorem on Complex Surface

Effective Divisor

Definition 5.3.2.1

Dimension of $\Omega^1(C)$

Hodge Theorem

De Rham Theorem

Riemann Inequality

Riemann-Roch Theorem

▶ ***Theorem*** ...

5.3.3 Applications

title

5.4 Hyperelliptic Riemann Surfaces

,

5.5 Projective Embedding

Chapter 6 Uniformization

6.1 Classification Theorem of Universal Covering

Chapter 7 Group Action on Riemann Surface

Chapter 8 Automorphisms

Definition 8.0.0.1 Define: For a Riemann surface X,

$$Aut(X) := \{f : X \to X | f \text{ is biholomorphic}\}$$

8.1 Automorphisms

8.1.1 Riemann Mapping Theorem: Simply Connected

▶ ***Theorem*** (Riemann Mapping Theorem) Every simply-connected surface is biholomorpic to one of the following:

- Riemann Sphere $\Sigma \cong^1 \mathbb{C}$;
- Complex Plane $\mathbb{C}$;
- Upper half plane $\mathbb{H}$;

∘ **Proof** ... □

And recall a small skill

♠ **Proposition 8.1.1.1** There exists biholomorphism

$$\mathbb{H} \cong \mathbb{D}$$

where $\mathbb{D}$ is the unit disk.

∘ **Proof** Given by Cayley transformation

$$\phi(z) := \frac{z-i}{z+i} \qquad \mathbb{H} \to \mathbb{D}$$

□

◇ **Lemma 8.1.1.2** $Aut(\mathbb{D}) = \{e^{i\theta}\frac{z-a}{1-\bar{a}z} | a \in \mathbb{C}\}$;

∘ **Proof** By Schwartz's lemma:

...

□

♠ **Proposition 8.1.1.3**

$$Aut(\Sigma) = \{z \mapsto \frac{\alpha z+\beta}{\gamma z+\delta} | \alpha\delta - \beta\gamma \neq 0, \alpha, \beta, \gamma, \delta \in \mathbb{C} \cup \{\infty\}\} \cong PGL(2, \mathbb{C})$$
$$Aut(\mathbb{C}) = \{z \mapsto az + b | a \neq 0\} \cong \mathbb{C} \rtimes_\rho \mathbb{C}^\times \qquad \rho :$$
$$Aut(\mathbb{H}) = \{z \mapsto \frac{\alpha z+\beta}{\gamma z+\delta} | \alpha, \beta, \delta, \gamma \in \mathbb{R} \cup \{\infty\}, \alpha\delta - \beta\gamma = 1\} \cong PSL(2, \mathbb{R})$$

- **Proof**
-
-
- We have discussed the necessary steps.

□

8.1.2 Group Action

▶ ***Theorem*** Let:

8.1.3 Finite Group Action on Riemann Surface

8.1.4 $\mathbb{Z}/p$-Action on Surfaces

8.2 Hurwitz's Theorem

8.2.1 Schwartz's Theorem

8.2.2 Hurwitz's Theorem

▶ ***Theorem*** (Hurwitz,1893, Automorphism Group of Riemann Surface with Genus ≥ 2)

- **Proof** ... □

Definition 8.2.2.1

- (Hurwitz Group);
- ($(2,3,7)$ Triangle Group)

▶ ***Theorem*** (Hurwitz's Criterion)

- **Proof** ... □

8.3 Fuchsian Groups

Definition 8.3.0.1

-
-

8.4 Hurwitz Grup

8.5 Compact Riemann Surface Theory

8.6 Non-Commpact Riemann Surface Theory

Chapter 9 Riemann Theta Function

Chapter 10 Jacobi Variety

Now, we just discuss the Jacobi Variety related to Riemann surface(in this part). For Jacobi Variety in general. For the Jacobi variety in general , we will do some note for Milne's note:

`https://www.jmilne.org/math/xnotes/JVs.pdf`

in a new part...

Chapter 11 Riemann-Hilbert Correspondence

Chapter 12 Differential Galois Theory (for Riemann Surface)

Chapter 13 Stein Manifold

We will research the Stein Manifold in higher dimension, in the a new note. Here, we just discuss the connection between Stein manifold and Riemann Surface.

Chapter 14 Mapping Class Group

14.1 Mapping Class Group and Dehn Twisted

We will research Riemann surface. Thus, we assume:

Every surface in this context, is orientable.

14.1.1 Preparation: Some of Concepts

Topology of Mappings

Topology: Whitney Topology

♠ **Proposition 14.1.1.1** If S is comapct, then in the category of topological space,

$$S_{Weak-Whitney} = S_{Strong-Whitney}$$

Some of Sets

Definition 14.1.1.2 Let S be a surface, and $P \subseteq S$:

- $$\mathrm{Diff}(S) := \{f : S \to S | f \text{ is diffeomorphism}\}$$
- $$\mathrm{Diff}(S, P) = \{f \in \mathrm{Diff}(S) | f|S = id_S\}$$
- Then, under the weak-whitney topology, denote
  - The identity component: $\mathrm{Diff}^\circ(S, P) \subseteq \mathrm{Diff}(S, P)$;
  - The orientation preserving diffeomorphism $Diff^+(S, P)$;

◇ **Lemma 14.1.1.3** Actually, ${}^0(S, P)$ is the pointwise isotopy subgroup(to identity) of (S, P), which is invariant on P.

∘ **Proof**

□

◇ **Lemma 14.1.1.4**

$$^\circ(S, P) \lhd^+ (S, P)$$

∘ **Proof** Firstly, $^\circ(S, P) \subseteq^+ (S, P)$: Notice that, if

Secondly, it is a normal subgroup: Under the operator of composition, it is actually a subgroup. And the topological subgroup is normal, because it is stable under conjugation. □

Thus, we could define

Definition 14.1.1.5 Let P be a smooth surface with $S \subseteq P$, then

$$Mod(S, P) =^{+} (S, P)/^{\circ}(S, P)$$

Remark In general, we will talk about

$$\text{Homeo}(S, P) \qquad \text{Homeo}^{+}(S, P) \qquad \text{Homeo}^{\circ}(S, P)$$

Then,

$$Mod(S) = \pi_0(Homeo^{+}(S)) = Hoemo^{+}(S)/Homeo^{\circ}(S)$$

14.1.2 Dehn Twists

Definition 14.1.2.1 Suppose S be an orientable surface, and

14.1.3 Fundamental Group

Definition 14.1.3.1 ...

14.2 Conformal Structure

14.3 Geometry of Upper Half Plane

14.4 Hyperbolic Surface

Chapter 15 Moduli Space of Riemann Surface

Chapter 16 Tecihmüller Space

Chapter 17 Tropical Curves

Chapter 18 Curves and Divisors

18.1 Algebraic Curve and Analytification

In this section, we will mainly talk about the smooth curve over $\mathbb{C}$, and the normalization.
Especially: Smooth Projective Curve and Compact Riemann Surface.

18.1.1 Analytification

18.1.2 Singular Curve

18.1.3 Hyperelliptic Curves

18.2 Divisors

18.3 the Riemann-Roch for Curve

18.3.1 Applications

18.4 Degree of Projective Curve

18.4.1 Poincare-Hopf Theorem

18.5 Abel's Theorem